Area 6: Supportive Knowledge

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

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6.A Manufacturing Methods

1. Key concepts

- Machining: spindle speed N = 12V/(πD) (V in sfm, D in in). Feed rate = f·N. Time = L/(fN). Taylor tool life $VT^n = C$.
- Process choice: casting (complex shapes, lower strength), forging (good grain flow, fatigue strength), machining (precision), additive (complex shapes, low volume), thread rolling (cold work, better fatigue).
- Heat treatment: annealing (soft), normalizing, quench and temper, case hardening (carburizing or nitriding).

2. Essential formulas

Name	Formula	Variables / units
Spindle speed	$N = 12V/(\pi D)$	
Cutting time	$t = L/(fN)$	
Taylor	$VT^n = C$	
MRR (turning)	$MRR = 12V \cdot f \cdot d$	in^3/min

3. Handbook search keywords

cutting speed | Taylor tool life | machining time | manufacturing processes

4. Worked examples

Example 1: Taylor tool life
n = 0.25, C = 400. Find T at V = 300 ft/min.

1. T = (400/300)^(1/0.25) = 3.16 min.

Answer: 3.16 min

5. Common traps

- Leaving out the 12 in N = 12V/πD.
- Inverting the exponent in Taylor (use 1/n).

6.A Manufacturing Methods: one-page cheat summary

► $N = 12V/\pi D$

► $t = L/fN$

► $T = (C/V)^{(1/n)}$

Formulas

Spindle speed	$N = 12V/(\pi D)$
Cutting time	$t = L/(fN)$
Taylor	$VT^n = C$
MRR (turning)	$MRR = 12V \cdot f \cdot d$

Search keywords

cutting speed, Taylor tool life, machining time, manufacturing processes

Top traps

- Leaving out the 12 in N = 12V/πD.
- Inverting the exponent in Taylor (use 1/n).

6.B Fits and Tolerances

1. Key concepts

- Hole-basis system: the hole lower limit equals basic size (H). Max clearance = largest hole − smallest shaft. Min clearance (or max interference) = smallest hole − largest shaft.
- ANSI B4.1 classes: RC running, LC locational clearance, LT transition, LN locational interference, FN force/shrink. ISO: H7/g6 sliding, H7/h6 locational, H7/p6 interference, H7/s6 drive.
- Tolerance stack: worst case $T = \Sigma T_i$; statistical (RSS) $T = \sqrt{\Sigma T_i^2}$.

2. Essential formulas

Name	Formula	Variables / units
Max clearance	$C_{max} = hole_{max} - shaft_{min}$	
Min clearance	$C_{min} = hole_{min} - shaft_{max}$ (negative = interference)	
Worst case	$T = \Sigma t_i$	
RSS	$T = \sqrt{\Sigma t_i^2}$	

3. Handbook search keywords

limits and fits · ANSI B4.1 · ISO 286 · tolerance stack-up RSS

4. Worked examples

Example 1: Stack-up
Four parts at ±0.002, ±0.003, ±0.002, ±0.004 in. Find the worst-case and RSS tolerances.

1. Worst case = ±0.011 in.
2. RSS = $\sqrt{(4 + 9 + 4 + 16)} \times 10^{-3}$ = ±0.0057 in.

Answer: ±0.011 (WC), ±0.0057 (RSS)

5. Common traps

- Swapping max and min clearance.
- Using RSS when the question asks for worst case.

6.B Fits and Tolerances: one-page cheat summary

- ▶ $C_{max} = H_{max} - s_{min}$
- ▶ $C_{min} = H_{min} - s_{max}$
- ▶ WC Σt; RSS $\sqrt{\Sigma t^2}$
- ▶ FN = force fit

Formulas

Max clearance	$C_{max} = hole_{max} - shaft_{min}$
Min clearance	$C_{min} = hole_{min} - shaft_{max}$ (negative = interference)
Worst case	$T = \Sigma t_i$
RSS	$T = \sqrt{\Sigma t_i^2}$

Search keywords

limits and fits, ANSI B4.1, ISO 286, tolerance stack-up RSS

Top traps

- Swapping max and min clearance.
- Using RSS when the question asks for worst case.

6.C Codes and Standards

1. Key concepts

- Know which code covers what:
- ASME BPVC Section VIII: unfired pressure vessels. Section I: power boilers. ASME B31.1: power piping. B31.3: process piping.
- ASME Y14.5: GD&T. Y14.100: drawing practice. AWS A2.4: weld symbols. AWS D1.1: structural steel welding.
- AGMA 2001: gear rating. ANSI/ABMA 9/11 and ISO 281: bearing ratings. SAE J429 and ASTM A325/A490: bolt grades. ASTM E8: tension test. ISO 286 and ANSI B4.1: fits.
- The exam may supply excerpts of specific standards. Check the last page of the NCEES MDM specification for the current list.

2. Essential formulas

Name	Formula	Variables / units
(no formulas)	Recognition topic. Know the scope of each code.	

3. Handbook search keywords

ASME Section VIII | B31.3 | Y14.5 | AWS D1.1 | AGMA | ABMA

4. Worked examples

Example 1: Identify the code
You are designing process piping in a refinery. Which code governs?

1. Process piping falls under ASME B31.3 (power-plant piping is B31.1).

Answer: ASME B31.3

5. Common traps

- Confusing B31.1 (power) with B31.3 (process).
- Confusing AWS A2.4 (symbols) with D1.1 (structural welding).

6.C Codes and Standards: one-page cheat summary

- ▶ BPVC VIII vessels
- ▶ B31.1 power / B31.3 process
- ▶ Y14.5 GD&T
- ▶ AWS D1.1 welding
- ▶ AGMA gears; ABMA bearings
- ▶ SAE J429 bolts

Formulas

(no formulas)	Recognition topic. Know the scope of each code.

Search keywords

ASME Section VIII, B31.3, Y14.5, AWS D1.1, AGMA, ABMA

Top traps

- Confusing B31.1 (power) with B31.3 (process).
- Confusing AWS A2.4 (symbols) with D1.1 (structural welding).

6.D Computational Methods

1. Key concepts

- Linear interpolation from property tables: $y = y_1 + (y_2 - y_1)(x - x_1)/(x_2 - x_1)$.
- Root finding: Newton–Raphson $x_1 = x_0 - f(x_0)/f'(x_0)$. Bisection is slower but always converges.
- Numerical integration: trapezoid; Simpson's 1/3 rule (even number of intervals) is exact for cubics.
- FEA: refine the mesh where gradients are high. Sharp re-entrant corners are singularities, so model the real fillet. Check reactions against the hand-calculated loads.

2. Essential formulas

Name	Formula	Variables / units
Interpolation	$y = y_1 + (y_2 - y_1)(x - x_1)/(x_2 - x_1)$	
Newton–Raphson	$x_{k+1} = x_k - f/f'$	
Simpson 1/3	$\int \approx (h/3)[f_0 + 4f_1 + 2f_2 + \ldots + f_n]$	
Trapezoid	$\int \approx (h/2)[f_0 + 2f_1 + \ldots + f_n]$	

3. Handbook search keywords

linear interpolation · Newton-Raphson · Simpson's rule · finite element

4. Worked examples

Example 1: Newton–Raphson
Find the root of $f(x) = x^3 - 2x - 5$ starting at $x_0 = 2$ (one iteration).

1. $f(2) = -1$; $f'(x) = 3x^2 - 2 = 10$.
2. $x_1 = 2 - (-1)/10 = 2.1$.

Answer: 2.1 (true root ≈ 2.0946)

5. Common traps

- Using Simpson's rule with an odd number of intervals.
- Inverting the slope in Newton's method.

6.D Computational Methods: one-page cheat summary

- ► Interp: $y_1 + \Delta y \cdot (x - x_1)/\Delta x$
- ► N–R: $x - f/f'$
- ► Simpson: $h/3[1\ 4\ 2\ 4\ \ldots 1]$
- ► FEA: singular corners do not converge

Formulas

Interpolation	$y = y_1 + (y_2 - y_1)(x - x_1)/(x_2 - x_1)$
Newton–Raphson	$x_{k+1} = x_k - f/f'$
Simpson 1/3	$\int \approx (h/3)[f_0 + 4f_1 + 2f_2 + \ldots + f_n]$
Trapezoid	$\int \approx (h/2)[f_0 + 2f_1 + \ldots + f_n]$

Search keywords

linear interpolation, Newton-Raphson, Simpson's rule, finite element

Top traps

- Using Simpson's rule with an odd number of intervals.
- Inverting the slope in Newton's method.

6.E Instrumentation, Testing, Inspection, and Quality

1. Key concepts

- Measurement: the instrument resolution should be about 1/10 of the tolerance. A micrometer reads 0.0001 in; a caliper reads 0.001 in.
- Strain gage: ε = (ΔR/R)/GF.
- Process capability: C_p = (USL − LSL)/6σ; C_pk = min(USL − μ, μ − LSL)/3σ. 1.33 or more is good.
- Control charts: $\bar{X}$ chart limits $\bar{X} \pm A_2\bar{R}$; R chart limits $D_3\bar{R}$ and $D_4\bar{R}$.
- NDT: UT and RT find subsurface flaws. Dye penetrant finds surface-breaking flaws only. Magnetic particle finds surface and near-surface flaws (ferromagnetic parts only).

2. Essential formulas

Name	Formula	Variables / units
Strain gage	$\varepsilon = (\Delta R/R)/GF$	
Cp	$C_p = (USL - LSL)/(6\sigma)$	
Cpk	$C_{pk} = \min(USL - \mu, \mu - LSL)/(3\sigma)$	
$\bar{X}$ chart	$UCL = \bar{\bar{X}} + A_2\bar{R}$	

3. Handbook search keywords

process capability Cpk · control chart · strain gage factor · nondestructive testing

4. Worked examples

Example 1: Cpk
Limits 25.00–25.10 mm, μ = 25.035, σ = 0.010. Find C_p and C_pk.

1. C_p = 0.10/0.06 = 1.67.
2. C_pk = min(0.065, 0.035)/0.03 = 1.17 (not centered).

Answer: C_p = 1.67, C_pk = 1.17

5. Common traps

- Using C_p when the process is off center.
- Expecting dye penetrant to find internal flaws.

6.E Instrumentation, Testing, Inspection, and Quality: one-page cheat summary

- ► $\varepsilon = (\Delta R/R)/GF$
- ► $C_{pk} = \min(...)/3\sigma$
- ► $UCL = \bar{\bar{X}} + A_2\bar{R}$
- ► UT/RT subsurface; PT surface only

Formulas

Strain gage	$\varepsilon = (\Delta R/R)/GF$
Cp	$C_p = (USL - LSL)/(6\sigma)$
Cpk	$C_{pk} = \min(USL - \mu, \mu - LSL)/(3\sigma)$
$\bar{\bar{X}}$ chart	$UCL = \bar{\bar{X}} + A_2\bar{R}$

Search keywords

process capability Cpk, control chart, strain gage factor, nondestructive testing

Top traps

- Using C_p when the process is off center.
- Expecting dye penetrant to find internal flaws.

6.F Chemical Processes (Corrosion, Coatings, Treatments)

1. Key concepts

- Galvanic corrosion: the more anodic metal corrodes (Mg, Zn, Al, steel, … Cu, stainless (passive), Ti, graphite). Avoid a small anode with a large cathode; isolate dissimilar metals.
- Protection: coatings, galvanizing (zinc as a sacrificial anode), cathodic protection, inhibitors, material selection.
- Hydrogen embrittlement: high-strength steels after plating or pickling. Bake after plating.
- Surface treatments: carburizing and nitriding (hard case), shot peening (compressive residual stress), anodizing (aluminum).

2. Essential formulas

Name	Formula	Variables / units
(mostly concept)	Galvanic series order; area ratio effect	

3. Handbook search keywords

galvanic series | corrosion protection | hydrogen embrittlement | surface treatment

4. Worked examples

Example 1: Galvanic pair
Aluminum rivets in a stainless panel in seawater. What happens and how do you fix it?

1. Aluminum is anodic and the rivet area is small: severe rivet corrosion.
2. Fix: use stainless rivets (large anode, small cathode), isolate the metals, or coat the cathode.

Answer: The rivets corrode; change the material or isolate

5. Common traps

- Coating only the anode (any holiday then corrodes fast). Coat the cathode, or both.

6.F Chemical Processes (Corrosion, Coatings, Treatments): one-page cheat summary

- ▶ Anode corrodes
- ▶ Small anode/large cathode = bad
- ▶ Zinc protects steel sacrificially
- ▶ Bake after plating

Formulas

(mostly concept)	Galvanic series order; area ratio effect

Search keywords

galvanic series, corrosion protection, hydrogen embrittlement, surface treatment

Top traps

- Coating only the anode (any holiday then corrodes fast). Coat the cathode, or both.